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ICABMET
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ST. DOMINIC COLLEGE OF ASIA
REVOLUTIONIZINGEDUCATION™

"2nd INTERNATIONAL CONFERENCE ON APPLIED SCIENCE, BUSINESS AND MANAGEMENT, ENGINEERING & TECHNOLOGY (ICABMET-2026)"

30th & 31st Mar,2026

Dubai, UAE.

Organized by

St. Dominic College of Asia, Philippines and
Confworld Educational Research and Development Association

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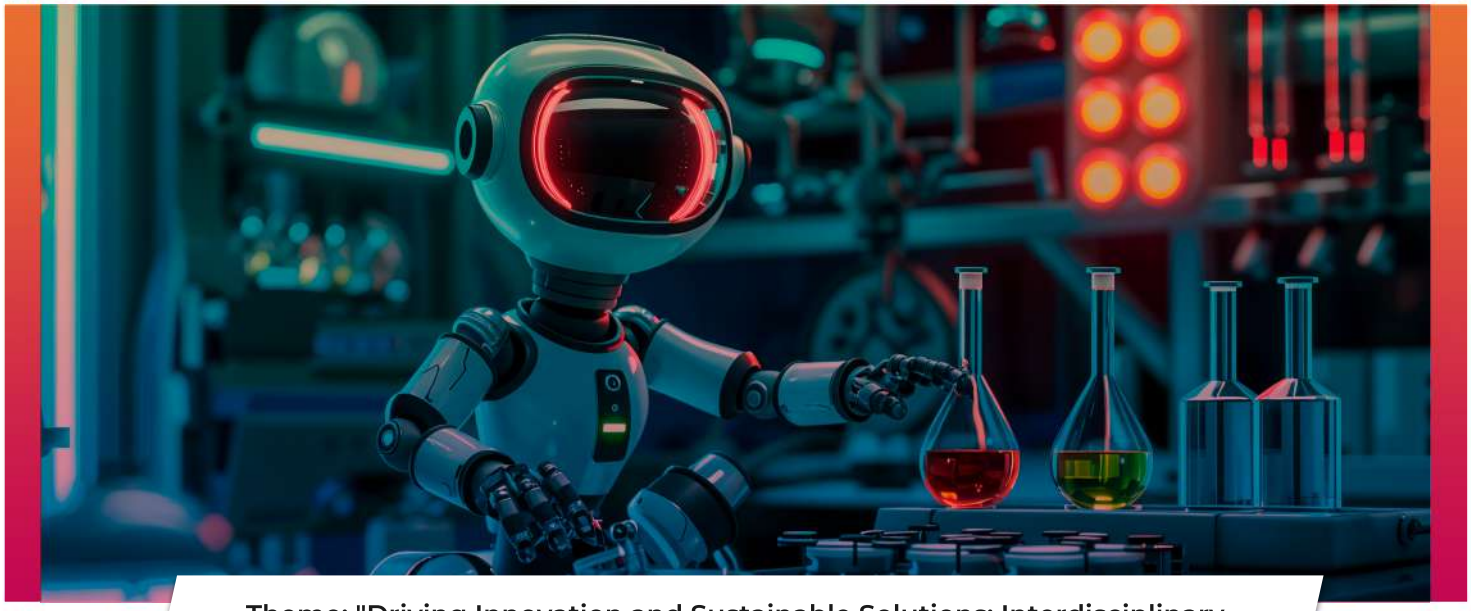
"2nd International Conference on Applied Science, Business and Management, Engineering & Technology (ICABMET-2026)"



Theme: "Driving Innovation and Sustainable Solutions: Interdisciplinary Approaches in Applied Sciences, Business and Technology"

30-31 Mar, 2026 | Dubai, UAE

ORGANIZED BY

St. Dominic College of Aia, Philippines and
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Theme:"Driving Innovation and Sustainable Solutions: Interdisciplinary Approaches in Applied Sciences, Business and Technology"

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Preface

ICABMET-2026 serves as a dynamic platform for fostering interdisciplinary collaboration and disseminating innovative research across the domains of applied science, business and management, engineering and technology. The conference features keynote lectures, workshops and research paper presentations that facilitate meaningful discussions on innovation, sustainability and global development.

Participants will have the opportunity to explore emerging trends, address contemporary challenges and examine significant advancements shaping these fields. The event promotes knowledge exchange, collaborative partnerships and professional networking while strengthening international academic and research connections. We warmly invite students, scholars, researchers and professionals to contribute and participate in this global academic forum. The proceedings of ICABMET-2026 present a collection of recent and impactful research contributions from diverse interdisciplinary domains, including Applied Science Innovations, Business and Management Trends, Engineering and Technological Advancements, Sustainable Development and Global Challenges, Healthcare Technology and Biomedical Engineering, Information Technology and Data Science, Education, Society and Innovation and Emerging Industrial Technologies.

All submitted manuscripts were subjected to a rigorous peer-review process and evaluated by two to four subject-matter experts to ensure high standards of academic quality, originality and relevance to the conference themes. The papers included in these proceedings reflect the contemporary research landscape and provide valuable insights that support continued scholarly exploration and technological advancement. This volume is intended to serve as a meaningful reference for academics, researchers and practitioners engaged in multidisciplinary research and innovation.

We sincerely acknowledge the support of partner universities, research institutions and academic collaborators whose contributions played a significant role in the success of this conference. Our special appreciation is extended to the dedicated reviewers and domain experts whose careful evaluations and scholarly commitment enhanced the overall quality of this publication.

We extend our sincere appreciation to the external reviewers for their valuable expertise and dedication throughout the peer-review process. Our heartfelt thanks also goes to all contributing authors whose research contributions enriched ICABMET-2026 with diverse perspectives and impactful insights.

Since September 2025, the Organizing Committees have received more than 50 manuscript submissions covering a broad range of interdisciplinary topics, including Applied Science Innovations, Business and Management Trends, Engineering and Technological Advancements, Sustainable Development and Global Challenges, Healthcare Technology and Biomedical Engineering, Information Technology and Data Science, Education, Society and Innovation.

We are grateful to all participants whose active involvement and scholarly contributions played a vital role in the success of ICABMET-2026. In particular, we thank our keynote speakers, session presenters and authors for their enthusiasm, commitment and academic excellence.
We also express our deep appreciation to the Technical Program Committee and review panel for their diligence and expertise in maintaining the academic quality of the conference. Finally, we acknowledge the Organizing Committee for their dedicated efforts and careful coordination in making the event successful.

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About ICABMET

ICABMET-2026 serves as an interdisciplinary platform where Engineering and Technology, Science, Education, Business, Social Sciences and Humanities converge to explore innovative solutions for a sustainable future. The event will take place on March 30–31, 2026 in Dubai, UAE, this international conference brings together distinguished researchers, scholars, academicians and industry professionals from around the world to share knowledge, exchange ideas and promote collaborative research.

The conference provides an engaging forum for presenting pioneering research, innovative concepts and emerging technologies aimed at addressing pressing global challenges. Through insightful keynote speeches, research presentations and interactive discussions, ICABMET-2026 encourages meaningful dialogue, networking opportunities and collaborative initiatives that drive innovation and societal progress.

Purpose of ICABMET

The "2nd International Conference on Applied Science, Business and Management, Engineering and Technology (ICABMET-2026)" serves as a premier platform dedicated to promoting sustainability and innovation across these critical fields. Guided by the theme "Driving Innovation and Sustainable Solutions: Interdisciplinary Approaches in Applied Sciences, Business and Technology," the conference seeks to address complex global challenges through collaborative and multidisciplinary perspectives.

ICABMET-2026 provides an opportunity for researchers, scholars and professionals to present innovative research, exchange knowledge and engage in meaningful discussions on emerging technologies and sustainable practices. By encouraging interdisciplinary collaboration and knowledge sharing, the conference aims to inspire solutions that contribute to global progress and sustainable development while expanding the academic and professional horizons of its participants.

Objectives of ICABMET

- To promote interdisciplinary research and constructive discussion among scholars and professionals in applied sciences, business and management, engineering and technology.
- To encourage the exploration and advancement of innovative and sustainable practices that address contemporary global challenges.
- To build a collaborative network of researchers, academicians and industry experts dedicated to developing practical and impactful solutions for real-world problems.

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About CERADA

The Confworld Educational Research and Development Association (CERADA) is a professional association dedicated to fostering Engineering, Science & Technology, Education, Business & Management, Social Sciences, Arts & Humanities and beyond by supporting research initiatives, disseminating cutting-edge insights, driving industry trends and fostering related endeavors.

CERADA aims to transform the process of innovation, collaboration and knowledge-sharing into a digital format by creating a unified virtual scientific community worldwide. This initiative will include research support, networking, educational opportunities, joint ventures, publications and other related activities.

CERADA is establishing strong academic, scientific and industry networks across Asia and the Middle East. Its presence extends to countries such as Indonesia, Malaysia, Philippines, Singapore, Turkey, Thailand and the United Arab Emirates (UAE).

Our Mission

Confworld Educational Research and Development Association (CERADA) advances global education and research through international conferences, research assistance and collaborative publications, fostering an inclusive environment for knowledge sharing and innovation.

Our Vision

Confworld Educational Research and Development Association (CERADA) envisions a borderless educational research community committed to lifelong learning and excellence, aiming to be a catalyst for transformation advancements through cutting-edge research and international partnerships.

What We Do

Confworld Educational Research and Development Association (CERADA) dedicated to building a dynamic community of professionals, educators, researchers and innovators in the realms of engineering, science & technology, business & management, social sciences & humanities and education. As part of this mission, we offer a platform that enables:

CERADA organizes a wide range of educational initiatives, which includes conferences, workshops, webinars, guest lectures, seminars, short-term training programs, public education programs and faculty development programs, all focused on Engineering and Science & Technology.

With a focus on curiosity, innovation and the latest trends in Engineering and Technology, CERADA is devoted to advancing knowledge in these fields.

We are committed to provide easy access to academic resources and support for aspiring students and research scholars from both urban and rural areas.

We are committed to fostering partnerships with universities, organizations and associations to promote knowledge sharing and work collectively toward building a better future.

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About SDCA

The St. Dominic College of Asia, also referred to by its acronym SDCA, is a private co-educational basic and higher education institution in Bacoor, Cavite, Philippines. It was founded by Don Gregorio and Doña Dominga Andaman in 2003 and initially named St. Dominic College of Arts & Sciences. SDCA offers in preschool, primary, secondary, tertiary, postgraduate education levels, as well as vocational education program.

In 2019 St. Dominic College of Asia has granted Deregulated Status by Commission on Higher Education (Philippines), the grant of deregulated status is given to private HEIs that have consistently shown exemplary performance in the provision of education, research and extension services, at the same time rationalizing supervision of private HEIs through progressive deregulation and last 2020, the Philippine Association of Colleges and Universities Commission on Accreditation (PACUCOA) awarded St. Dominic College of Asia as the TOP 3 College with Highest Accredited Programs in the Philippines for the third time.

St. Dominic College of Asia is divided into five schools, graduate school, basic and senior high school education department and SDCA offers 17 Undergraduate degree programs and 3 graduate degree programs. In 2022, Dr. Andaman announced the plan to build the School of Medicine for the 20th anniversary of SDCA.

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MD's Message, CERADA

Dr. Geethalakshmi R
MD & Founder,
Confworld Educational Research and
Development Association (CERADA)

On behalf of CERADA and the Organizing Committee, I would like to express my sincere appreciation to all participants, keynote speakers, delegates, reviewers and researchers whose valuable contributions made ICABMET-2026 a highly successful and memorable event.

It is my great pleasure and honor to warmly welcome you all to the International Conference on Applied Science, Business and Management, Engineering and Technology (ICABMET-2026). We are delighted to host this prestigious global gathering here in Dubai, UAE, on March 30–31, 2026. ICABMET-2026 has been organized with the vision of creating a dynamic platform that brings together experts, scholars and practitioners from diverse disciplines including applied sciences, business and management, engineering, technology, education and social sciences. The conference aims to encourage meaningful dialogue, share innovative research and explore interdisciplinary solutions to address the complex challenges faced by our world today.

This year's conference theme, "Driving Innovation and Sustainable Solutions: Interdisciplinary Approaches in Applied Sciences, Business and Technology," highlights the importance of collaboration across fields in shaping a sustainable and technologically advanced future. Through keynote presentations, research paper sessions, and interactive discussions, ICABMET-2026 offers a valuable opportunity for participants to exchange ideas, discover new perspectives and build lasting professioal connections.

We are truly honored to welcome distinguished keynote speakers and leading researchers from around the world who will share their knowledge, insights and experiences. Their contributions will undoubtedly enrich our discussions and inspire innovative thinking.

I would also like to express my sincere gratitude to the organizing committee, technical program committee, reviewers, partner institutions and volunteers whose dedication and hard work have made this conference possible. My heartfelt thanks to all authors and participants whose valuable research contributions form the foundation of this event.

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Conference Chair

Dr. Diwakar Ramanuj Tripathi
HOD, Dept. of Computer Science,
S.S. Maniar College of computer and Management,
R.T.M. Nagpur University, Nagpur, Maharashtra, India.

Dr. Diwakar Ramanuj Tripathi (MCA, Ph.D., LL.B., FIARA, FSASS, FISROSET) is currently serving as the Head of the Department of Computer Science and Controller of Examinations at S.S. Maniar College of Computer & Management, Nagpur, Maharashtra, India. He is also recognized as a Ph.D. supervisor at Rashtrasant Tukadoji Maharaj Nagpur University and a visiting faculty member at ISBM University.

With over 13 years of professional experience in the field of Information Technology, Dr. Tripathi brings extensive expertise in consulting, training, and IT management. His core competencies include capability enhancement, enterprise analysis, and IT security and business analysis. He is an ISO-certified Lead Auditor and a Microsoft Certified Technology Specialist. Dr.Tripathi has contributed significantly to academic research, having published more than 20 research papers in reputed international and national journals and conferences, many of which are Scopus-indexed and UGC-approved. He holds over 10 national and international patents and has registered 11 copyrights. Additionally, he is a registered trademark attorney with the Government of India.

He has been the recipient of over 12 national and international awards in recognition of his contributions to the field of computer science. Notable accolades include the Bhart Nepal Shiksha Vibhushan Award, the Innovative Research Award, and the Albert Einstein Award at the international level.

Dr. Tripathi holds fellow memberships in several esteemed professional organizations, including:

- Institute of Research Engineers and Doctors (IRED), New York, USA
- International Scientific Research Organization for Science, Engineering and Technology (ISROSET)
- Indian Academicians and Researchers Association (IARA)
- Scholars Academic and Scientific Society (SAS Society)

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Keynote Speaker

Dr. Rania Lampou
Ministry of Education & Regulatory Affairs,
Directorate of Educational Technology & Innovation, Greece.

Rania Lampou is a multi-awarded Global educator, STEM instructor, ICT teacher trainer, researcher, international speaker, author, and global peace ambassador in Greece. Currently, she is working at the Greek Ministry of Education, at the Directorate of Educational Technology and Innovation, where she writes STE(A)M projects for Greek schools. She is also a STEM instructor at the Greek Astronomy and Space Company (Annex of Salamis).

She has been awarded many national and international prizes (over 200) and she is a "Global Teacher Award 2019, 2020 & 2021" (AKS Awards) winner and a "Global Teacher Prize finalist 2019" (Varkey Foundation). She has been selected as "Global Icon 2020" featured in "Passion Vista" Magazine, among the top 10 women entrepreneurs featured in "Fortune", among the "100 most successful women in business" featured in an Amazon book by "Global Trade Chamber" and among the "Top 10 CEO's of the year 2021-22" by "Digisharks" and "Indian Alert". She also features in the books: "30 Shining stars of the world", "Heidens Book 2023", "Queens Book 2021 & 2022", and "Hoinser Global Women Book 2023" by "Hoinser Media Group".

She promotes STEM vision by introducing STEM in astronomy and physics projects and combines STEM with language teaching. She is the founder and international coordinator of many innovative international projects that focus on the United Nations' Sustainable Development Goals. She is also an author of scientific books for kids.

She is a social activist, and a member of the educational committee of the Ecumenical Delphic Union in collaboration with Harvard University. She collaborates with many international organisations and she has received many humanitarian and peace awards from International Humanitarian Organizations.

She is an Editorial Board Member of many journals. She has presented her research in numerous international conferences and she has published work in various journals.

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Keynote Speaker

Dr. Manar Alkhatib
Head of Program BSc CS (AI/CyS),
Assistant Professor,
The British University in Dubai,
Dubai, UAE.

Dr. Manar earned her PhD in Computer Science from The British University in Dubai in July 2019. During her graduate studies, she worked as Faculty to teach various courses and had the opportunity to develop her skills as an educator at British University in Dubai, Zayed University, and HCT.

She has publications on Deep Neural Network, machine translation, sentiment analysis, computational morphology, data mining and information retrieval. Her research interests in Artificial Intelligence include Large Language Model, Deep Learning, Machine Learning, Big Data Analysis, Natural Language Processing (NLP), Computational Linguistics (CL), Image processing and Speech Recognition.

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Keynote Speaker

Dr. Renante A. Diamante
Dean,
College of Information and Communications Technology (CICT),
Iloilo State University of Fisheries Science and Technology (ISUFST),
Philippines.

Dr. Renante A. Diamante is a distinguished figure in the field of IT education. He currently serves as the Dean of the College of Information and Communications Technology (CICT) at Iloilo State University of Fisheries Science and Technology (ISUFST), Main Campus Poblacion Site, Barotac Nuevo, Iloilo. Additionally, he holds the position of Associate Professor IV and actively contributes to the university's graduate program as Dean and a Professor as well (Master in Information Technology and Doctor of Philosophy major in Educational Technology program).

Dr. Diamante's dedication to education extends beyond his current role. He previously served as the President of the Local Teacher and Employee Association (ISCOFCBN-TEA) from 2019 to 2022 and 2024 to present and held leadership positions as Dean of Iloilo State College of Fisheries – Barotac Nuevo Campus in 2014 – 2016).

He also served as Chief Operating Officer/Campus Administrator at System Technology Institute-Center of Education (2004-2007) at Barotac Viejo, and Dean of Cabalum Western College (2009-2011). He also has extensive experience as a full-time IT faculty member at Saint Paul University of Iloilo System (2001-2004). He is also a curriculum consultant and a part-time faculty at ABBA Institute of Technology, Inc. (2001-2004).

Dr. Diamante's academic credentials are equally impressive. He holds a Doctor of Information Technology (2020), a Master of Arts in Teaching major in Information and Communications Technology (2013), a Master of Information Technology (2015), and a Bachelor of Science in Computer Science (2001). A Two-Year Computer Technician (1997). He received the K to 12 Graduate Scholarship Program sponsored by the Commission on Higher Education (CHED) from 2016 to 2020. His passion for research is evident in his authorship of numerous scientific and developmental research papers presented at international and local conferences and published in both refereed and Scopus-indexed journals. He was invited as a Keynote Speaker and a peer reviewer at International research conferences. His research interests lie in data mining, image processing, and machine learning, demonstrating his commitment to pushing the boundaries of the IT field. He has also authored several books on Research (Quantitative and Method of Research). Dr. Diamante is a multifaceted individual who combines extensive experience in leadership, education, and research, making him a valuable asset to ISUFST and the broader IT education landscape.

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Keynote Speaker

Asst. Prof. Dr. Algene Malte De Guzman, MAELT, TESOL
Chief of the Center for Instructional Materials Development and Assessment,
Polytechnic University of the Philippines,
Philippines.

Asst. Prof. Dr.Algene Malte De Guzman is a highly accomplished educator, researcher and academic leader at the Polytechnic University of the Philippines (PUP). He currently serves as University Chief for the Center for Instructional Materials Development and Assessment, where he spearheads faculty training in module writing, chairs the University Evaluation Committee, and designs evaluation guidelines to ensure the rigor and validity of instructional materials across the PUP system. Alongside this role, he lectures at the Graduate School and the College of Arts and Letters, handling advanced courses in grammar, research methodology, stylistics, translation studies and technical writing.

With more than two decades of professional experience, Dr. De Guzman has held leadership and teaching positions in premier institutions such as West Bay College, San Beda College Alabang, Malayan Colleges Laguna, and iACADEMY. His teaching portfolio spans a wide range of disciplines from literature and humanities to workplace communication and foreign languages demonstrating his versatility and breadth of expertise. He has also authored institutional manuals on research and innovation and actively participated in academic councils and executive committees, contributing to institutional growth and development.

A prolific author, Chief De Guzman has co-written and coordinated more than a dozen of text-books, including Delving Discourses (2020), Processing Personal Progress (2020), Empowering the 21st Century Communicators (2019), Purposive Communications for the 21st Century Communicators (2018), and Maximizing Macroskills (2015). These works have become valuable references for students and educators in communication and language studies. His contributions extend beyond writing, as he has been a plenary and resource speaker in numerous national and international conferences, delivering lectures on instructional material development, effective communication and research writing.

His excellence has been recognized through numerous awards, including Most Outstanding Professional in Training and Development (2025), Most Outstanding Resource Speaker of the Year (2024), and Global Educator and Trainer of the Year (2023, Vietnam). He has also received international citations such as Best Faculty Award – Male Overseas (India, 2022) and Humanities, Social Sciences, and Languages Educator of the Year (Greece, 2021). These distinctions affirm his stature as a global thought leader in education and training.

Beyond academia, Dr. De Guzman is deeply engaged in advocacies promoting workplace inclusivity, women's empowerment, and cancer awareness. He is a lifetime member of the Linguistic Society of the Philippines, the Philippine Association for Forensic and Legal Linguistics, and the Royal Institute of Linguists (Singapore), among many others. His career reflects a commitment to empowering learners and educators alike, making him a respected voice in both national and international academic communities.

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Keynote Speaker

Dr. Gerald Garcia Ocan
Dean,
Institute of Management and Business Allied Professions,
Paranaque City College, Philippines.

Gerald Ocan was born on October 24, 1974, in Rosario, Cavite, to his proud parents, Bayani and Priscila Ocan. He was raised in a humble household together with his four siblings: Jingle Ocan Aruta, Glen Ocan, Janice Ocan, and Josephine Ocan (deceased).

He completed his primary education at Agustin Abadilla Elementary School and his secondary education at Rosario National High School (formerly Agustin Abadilla High School), where he graduated as First Honorable Mention of his batch—an early reflection of his dedication and academic excellence.

Coming from a financially challenged family, Gerald learned the value of perseverance, discipline and resilience at a young age. Determined to finish his studies, he supported himself through buying and selling various goods while pursuing his education. His hard work paid off as he earned his Bachelor of Science in Business Administration, major in Marketing Management, from San Sebastian College Recoletos de Cavite. Without pause, he continued his journey by completing his Master in Business Administration at the same institution while working full-time. In December 2019, he reached another milestone by earning his Doctorate in Business Administration from the Polytechnic University of the Philippines proof that persistence and passion can overcome any limitation.

Gerald dedicated 19 years of service to GSIS Family Bank, a Government-Owned and Controlled Corporation (GOCC), where he held key roles in branch operations, loan operations, and human resource development. Throughout his corporate career, he remained committed to education, serving as a part-time professor in various colleges and universities in Manila and sharing his real-world experience with aspiring professionals.

On May 16, 2019, he joined St. Dominic College of Asia as a full-time faculty member under the Business Administration Program. His competence and leadership quickly became evident, leading to his promotion as Program Chairperson of the BS Business Administration program after only two consecutive semesters. A year later, he was appointed as Dean of the School of Business and Computer Studies, while concurrently serving as Program Head for both the Business Administration and Master in Business Administration programs. In May 2023, following an institutional reorganization and program clustering, he was appointed as Dean of the School of Graduate Studies, further expanding his leadership role while retaining his responsibilities as Program Head of the Business Administration and MBA programs.

Currently, Gerald serves as the Dean of the Institute of Management and Business Allied Professions at Parañaque City College and part time professor in different colleges and universities for both undergraduate and graduate studies, where he continues to inspire students, faculty and future leaders through academic excellence, ethical leadership and service.

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Session Speaker

Assoc. Prof. Dr. Mohd Zaki Bin Awang Chek
Associate Professor, Center for Actuarial Studies,
Faculty of Computer and Mathematical Sciences,
Universiti Teknologi MARA (UiTM),
Malaysia.

Associate Professor Dr. Mohd Zaki Awang Chek is a distinguished academic and researcher in the field of actuarial science, currently serving at the Actuarial Science Department, Faculty of Computer and Mathematical Sciences, Universiti Teknologi MARA (UiTM), Perak Branch, Malaysia. With over a decade of experience in higher education, he has established himself as a leading expert in actuarial digital learning, data analytics, and social security systems.

Dr. Zaki holds a Doctor of Business Administration (Actuarial Management) from Universiti Teknologi MARA (2017), complemented by a Master of Quantitative Sciences (Financial Engineering) (2009) and a Bachelor of Actuarial Science (2007) from the same institution. His academic foundation reflects a strong integration of actuarial science, financial engineering, and applied analytics.

Since joining UiTM in 2009 as a lecturer, Dr. Zaki has progressed through the ranks to Senior Lecturer (2017–2022) and subsequently Associate Professor (2023–present) . Throughout his career, he has taught extensively at undergraduate, postgraduate, and doctoral levels, covering a wide spectrum of subjects including actuarial mathematics, financial mathematics, pension mathematics, risk management, insurance analytics, and business data analytics. His teaching excellence is consistently reflected in outstanding student evaluation scores and his ability to integrate technology-enhanced learning approaches.

Dr. Zaki's research portfolio is highly impactful, focusing on social protection systems, actuarial modelling, and data-driven analytics. He has led and contributed to numerous funded research projects, including collaborations with SOCSO, Khazanah Research Institute, MCMC, and international bodies such as the London Mathematical Society . His research addresses critical national issues such as pension sustainability, employment injury schemes, commuting accidents and healthcare expenditure.

A prolific scholar, Dr. Zaki has published extensively in Scopus- and ERA-indexed journals, with research contributions spanning actuarial science, social security, financial planning and digital education. His recent works emphasize machine learning applications, learning analytics and micro-credential frameworks in actuarial education and social protection systems. He is also actively involved in academic publishing through journal articles, conference proceedings and book chapters.

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Session Speaker

Dr. Nagarajan S.
Assistant Professor (Former HoD & Associate Dean),
Department of Chemistry,
National Institute of Technology Manipur, Langol, Imphal,
India.

Dr. Nagarajan S. is a faculty member in the Department of Chemistry at the National Institute of Technology (NIT) Manipur. He previously served as a postdoctoral fellow at the University of Heidelberg, Germany, and as a visiting scientist at the University of California, Berkeley, USA.

He has delivered over 115 invited talks at various national and international conferences, reflecting his extensive academic engagement and expertise. His research interests include the application of artificial intelligence tools in modern research, along with contributions to areas such as sustainable development goals (SDGs), pollution control, materials science and polymer research.

Dr. Nagarajan has also conducted multiple workshops on AI tools for research, providing hands-on training to faculty members and research scholars. He is keen to contribute as an invited speaker and share his expertise through lectures and interactive sessions organized by esteemed institutions.

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Session Speaker

Dr. Anu Sayal
Associate Professor,
Centre for Artificial Intelligence,
Madhav Institute of Technology and Science,
Gwalior, Madhya Pradesh, India.

Dr. Anu Sayal is currently serving as Associate Professor at MITS Deemed University, Gwalior, India. She has worked as a Senior Lecturer in Mathematics/Statistics at Taylor's University, Malaysia, which is ranked among the top 30 in Asia and 253 globally.

She is also an accredited trainer by HRD Malaysia. With over 18 years of teaching and research experience, Dr. Sayal holds a Bachelor's degree in Non-Medical and a Master's degree in Mathematics from Guru Nanak Dev University, Amritsar. She earned her PhD in Mathematics from Uttarakhand Technical University, Dehradun.

Dr. Sayal has received several prestigious awards, including the Young Scientist Award in Mathematics, Statistics, and Computer Science from the Uttarakhand Council for Science and Technology, Dehradun, in February 2020. She was also honored with the Teacher of the Year 2020 Award by Divya Himgiri, in association with the Uttarakhand Council for Science & Technology, Government of Uttarakhand, and the Commission for Scientific & Technical Terminology, MHRD, Govt. of India. In 2021, she received the Guruwarya Samman Award from Harvest Educational Transformational Solutions (HETS) and the Best Woman Scientist Award from the Novel Research Academy. In January 2022, she was named Best Teacher by Harvest Educational Transformational Solutions. She also received the Academic excellence award, international faculty award on 3rd February 2025 from Novel Research Academy, Puducherry, India and Women faculty of the year award, Women power India awards 2025 on 8th March 2025, International Women's Day from MIT Square welfare foundation, India.

Dr. Sayal has authored numerous research papers published in renowned journals, including those indexed by Scopus and WOS, and has presented her work at national and international conferences. She has been a keynote speaker at various international events and has served as a resource person for numerous workshops, seminars and FDPs. In addition, she is an active reviewer and editorial board member for several international journals indexed by WOS and Scopus. Dr. Sayal has also organized a variety of workshops and Faculty Development Programs (FDPs).
Her research interests include Optimization, Inventory and Supply Chain Management, Fuzzy inventory models, Fuzzy Mathematics and Statistics.

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Session Chairs

Dr. Kelvin Lee Yong Ming
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Programme Director,
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Taylor's University, Malaysia.

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Dr. P. Sri Ram Chandra
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The Influence of Extrinsic Motivation, Organizational Culture, Perceived Organizational Support and Job Satisfaction on School Teachers' Innovative Performance in the Kingdom of Saudi Arabia

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Abstract

This study investigates the influence of extrinsic motivation, organizational culture and perceived organizational support on school teachers' innovative performance in the Tabuk region of the Kingdom of Saudi Arabia (KSA), with job satisfaction examined as a mediating factor. The aim of the study is to explore how organizational and motivational antecedents contribute to teachers' innovation in line with the country's Vision 2030 educational reforms. A quantitative research design was employed, utilizing structured survey questionnaires distributed to a stratified sample of 391 school teachers. Demographic information was analyzed using SPSS, while Structural Equation Modeling (SEM) was applied to test both direct and indirect relationships among constructs. The findings produced mixed outcomes regarding the hypothesized relationships. First, extrinsic motivation and perceived organizational support did not show significant direct effects on innovative performance, whereas organizational culture had a significant positive impact. Second, regarding job satisfaction, extrinsic motivation and perceived organizational support significantly enhanced job satisfaction, while organizational culture did not exert a meaningful effect. Third, job satisfaction itself exhibited a strong and significant positive impact on innovative performance, reaffirming its central role in linking organizational factors to innovation. With respect to mediation analysis, job satisfaction was found to significantly mediate the relationships between extrinsic motivation and innovative performance as well as between perceived organizational support and innovative performance. However, the mediating effect of job satisfaction in the relationship between organizational culture and innovative performance was not supported.

These results underscore that job satisfaction operates as a critical pathway for some, but not all, organizational antecedents of teacher innovation. The study contributes theoretically by refining social exchange theory and motivation-based frameworks in the educational context. Specifically, it demonstrates that extrinsic motivation and perceived organizational support indirectly foster innovative performance by enhancing job satisfaction, while organizational culture exerts its influence more directly.

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Methodologically, the use of SEM with bootstrapping adds rigor by disentangling supported and unsupported mediation pathways, providing nuanced insights into the mechanisms driving innovation. The study contributes theoretically by refining social exchange theory and motivation-based frameworks in the educational context. Practical implications suggest that policymakers and school leaders in KSA should prioritize strategies that improve teachers' perceptions of organizational support and strengthen extrinsic motivational mechanisms, such as recognition systems, fair policies, and professional development opportunities. These practices enhance job satisfaction, which subsequently encourages innovation. However, since organizational culture did not influence innovation indirectly through satisfaction, leaders should focus on developing cultural attributes that directly promote creativity, collaboration, and knowledge sharing. Despite these valuable insights, the study has limitations. The cross-sectional design restricts causal interpretations, and reliance on self-reported data may introduce bias. Additionally, the sample was limited to teachers in Tabuk, potentially reducing generalizability across other regions and educational systems. Future research should adopt longitudinal and multi-source approaches, explore alternative mediators such as engagement or empowerment and test the model across diverse cultural contexts.

Keywords: *Extrinsic Motivation, Organizational Culture, Perceived Organizational Support, School Teachers' Innovative Performance, Tabuk Region, Quantitative Study*

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Toward Sustainable Workplaces: Linking Green Working Conditions, Green Satisfaction and Turnover Intention

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Abstract

Employee satisfaction and retention depend on sustainable work environments which create positive work conditions. The study investigates how Green Working Conditions (GWC), Green Satisfaction (GS) and Turnover Intention (TI) interact through GS as a mediating factor. The researchers used Job Demands-Resources (JD-R) theory to conduct their quantitative research which examined 339 female workers in Bangladesh's ready-made garment (RMG) industry through self-administered surveys that used random sampling. The research findings demonstrate that GWC has a statistically significant negative effect on TI because better green working conditions lead to lower employee turnover intentions. Green Satisfaction worked as a significant mediator because it demonstrated a positive link with GWC and a negative link with TI. The results demonstrate that green HRM practices lead to higher employee satisfaction, which results in better employee retention through the development of green knowledge sharing and group cohesion and work engagement. The model, which has been validated through empirical research, demonstrates how organizations can implement sustainable human resource management and develop green practices which lead to employee retention and environmental protection. The study recommends interdisciplinary research to enhance understanding of the topic and its practical applications.

Keywords: *Green Working Conditions, Green Satisfaction, Turnover Intention, SEM*

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Gronwall-Bellman, Lyapnove Type Inequalities and Application to Fractional Differential Equations

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Abstract

Fractional integral inequalities are key tools in fractional calculus, providing important bounds and relationships for fractional integrals, which extend traditional integration to non-integer orders. These inequalities, such as those involving Hölder's, Hermite-Hadamard's, Ostrowski's, Hardy's, Gr¨uss-type, Jensen's Grönwall's, and Minkowski's integral inequalities, as well as generalized fractional integral inequalities are used to analyze the properties of fractional operators and fractional differential equations. They have wide applications in various fields, including applied mathematics, physics, and engineering particularly in modeling systems. By utilizing fractional integral inequalities, researchers can derive approximations, existence, uniqueness, stability of solutions, and improve the accuracy of models, making them essential in the study of complex phenomena and special functions. During this presentation, we will discuss the integral inequalities with generalized fractional integral and differential operators, as well as their application in a Gronwall-Bellman and Lyapunov type inequality to determine the necessary condition for the existence of non-trivial solutions for a boundary value problem with fractional differential equations.

Keywords: *Gronwall-Bellman Type Inequality, Lyapunov Type Inequality, Fractional Operator*

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